

Analytic Projective Geometry

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1 Introduction

Projective geometry is a new way of looking at geometry. We add a new perspective on looking at points in the plane, and see that this new perspective gives a lot of powerful results. Projective geometry is particularly close at heart to me because it is a topic that was introduced to me as a student at PROMYS and it's something that I decided to work out myself. The following are the results and intuition that I came to, and I hope that they can provide some insight into this elegant theory.

2 Points in Projective Geometry

Most introductions I have found for projective geometry have been motivated by things like introducing points at infinity or looking at what properties are preserved under projective transformations (think of perspective drawings). Although these questions will be answered, at least in part, later on, my understanding of projective geometry starts with a seemingly unmotivated idea.

Imagine you are standing in a room and looking up. If you tilt your head, you can see every point on the ceiling, given that there are no obstructions. The idea of two dimensional projective geometry is to replace your 2D plane with these "lines of vision." More specifically, we imagine that the 2D plane exists in $\mathbb{R}^3$ as the plane $z = 1$. The points on the plane are identified with the lines through the three dimensional origin going through that point. For example, the point $(1, 2) \in \mathbb{R}^2$ is identified with the line going through $(0, 0, 0)$ and $(1, 2, 1)$ (the third coordinate is 1 since the point exists on the plane $z = 1$). This is analogous with the earlier picture of looking at the ceiling where your head is at the origin, $z = 1$ is the "ceiling," and the lines through the origin are your "lines of vision."

Now, there are a few technicalities. Firstly, a "line of vision" would only be in the top half space of $\mathbb{R}^3$ since you only have eyes in the front of your head. Hence to imagine the full line through the origin as a "line of vision," we change our picture slightly. Instead of looking directly up at the ceiling, imagine you had a glass hemisphere around you (this will correspond to the unit sphere in $\mathbb{R}^3$). If the glass is perfectly see-through, then marking a point on the glass with a marker would look identical to marking a point on the ceiling. Now imagine that you complete the sphere and have a perfectly reflective mirror hemisphere below you (with glass still on top). Just for fun, imagine you're also wearing an invisibility cloak so that light can pass right through you. Now, if you make a mark on the mirror, what will you see? Well, the light would go through a point on the glass, reflect off the mirror and hit your eye. So if you look down at the mirror, you would see the point directly behind you, hence completing the "line of vision." Now why did we make it a mirror instead of just having a perfectly reflective floor? Well it turns out that the physical law of reflection actually makes this description work with what we intend lines of vision to mean (think of why it works with spheres and not with a flat floor below you).

Now what if you look horizontal, like perfectly horizontal? You wouldn't see a point on the ceiling and in more formal writing, your line of vision would be in the plane $z = 0$ which will never intersect with the plane $z = 1$ since they're parallel. So what do we make of this? Well if you look at a point

on the ceiling and gradually tilt your head down, you will be looking at a point on the ceiling further and further away. Hence, looking horizontally is sort of like “looking infinitely far away.” These lines actually correspond to so-called points at infinity. Now, how do points at infinity fit in with our sphere picture? Well, we want polar opposite horizontal lines of vision to correspond to the same point (since they’re the same line through the origin). The physical interpretation kind of fails us here because it’s where the glass hemisphere meets the mirror hemisphere. You can imagine there’s some wonky cosmic glue here if you so desire. We will come back to what these points at infinity correspond to later.

These lines through the origin corresponding to points on $\mathbb{R}^2$ and points at infinity together make what’s called the real projective plane $\mathbb{R}P^2$ (2 because we’re using 3D to talk about something 2 dimensional). The way to mathematically describe them is to use what are called projective coordinates. These are just coordinates that represent some point on your line of vision and all points on the same line are considered the same. They are often written as $[x : y : z]$ with the 3 coordinates reminding you it’s 3 dimensional, and the $:$ in between reminds you that the coordinates sort of act as ratios between the coordinates. For example, $[1 : 2 : 1]$, $[-2 : -4 : -2]$, $[9 : 18 : 9]$ all lie on the line through the origin that goes through the point $(1, 2, 1)$. Hence, we consider all of these points to be the same projectively and write $[1 : 2 : 1] = [-2 : -4 : -2] = [9 : 18 : 9]$. It makes sense to put equalities between them because really, they’re all serving to represent the point $(1, 2)$ in $\mathbb{R}^2$.

Say our z -coordinate of some point in projective space is nonzero so that we have $[x_0 : y_0 : z_0]$. Then, by rescaling, we have $[x_0/z_0 : y_0/z_0 : 1]$ which is a point on the $z = 1$ plane. Informally, if you “look” at an object at (x_0, y_0, z_0) from the origin, it would be indistinguishable from looking at an object at $(x_0/z_0, y_0/z_0, 1)$. Points at infinity lie on the plane $z = 0$ and so have projective coordinates with z -coordinate as 0. For example, $[1 : 2 : 0]$, $[3 : 7 : 0]$ are two distinct points at infinity. The term point at infinity makes sense because if we naively try to divide by the z -coordinate, we would get numbers divided by 0 which is informally infinite. Assuming the x -coordinate for a point at infinity is nonzero, we can divide it to get that $[x_0 : y_0 : 0] = [1 : y_0/x_0 : 0]$. The final expression has some y -coordinate divided by some x -coordinate, which is something that is similar to the formula for a slope. It turns out that points at infinity are closely related to the slopes that we can see in this way.

3 Lines

Let’s now discuss what lines look like in projective space. I will be referring both to lines in $\mathbb{R}^2$ (on the plane $z = 1$) and lines in $\mathbb{R}^3$ (which are projected onto points). To try and avoid confusion, I will denote objects on the plane with lowercase letters like p, ℓ and objects in 3D that project onto points or lines in the plane with capital letters corresponding to the 2D object like P, L .

We have just seen that a point p in the plane corresponds to a line through the origin in $\mathbb{R}^3$. Now what does a line ℓ look like in $\mathbb{R}^3$? Well, ℓ is composed of many points p and each of those points correspond to lines through the origin, P . Now, ℓ in $\mathbb{R}^3$ corresponds to a collection of many such lines through the origin. With some imagination, it can be seen that this should be a plane through the origin. The equation for a generic plane through the origin is $\alpha x + \beta y + \gamma z = 0$. If you haven’t seen this before, think of lines through the origin in 2D being written as $\alpha x + \beta y = 0$. What should be the equation of the plane L corresponding to the line ℓ ? We know that the intersection of the plane L with the plane $z = 1$ (which we are considering to be our 2 dimensional plane) should be the equation of our line in two dimensions. The intersection of the two planes would correspond to all (x, y, z) triples that satisfy $\alpha x + \beta y + \gamma z = 0$ (L) and $z = 1$ (2D plane). Now, we can simplify a little bit by substituting $z = 1$ into our equation for L : $\alpha x + \beta y + \gamma = 0$, which is the line in the plane. For example, if our line was $3x + 4y + 5 = 0$, the corresponding plane would be $3x + 4y + 5z = 0$. This idea of throwing z ’s into our equation is called homogenizing the equation.

Now, there are many ways to write the equation of a line (e.g. slope-intercept, slope-point) but it will turn out later to be convenient to use lines in standard form: $ax + by + c = 0$. As motivation for this, one common theme that will come up later is that projective geometry allows us to talk about things

with great generality. With slope-point or slope-intercept form, it isn't possible to write the equations of vertical lines. But with standard form, it is very easy (e.g. $x - 3 = 0$). More importantly, however, the next section will show that standard form is a very useful tool when discussing dual properties of points and lines.

There is one line in $\mathbb{R}P^2$ which is distinguished from lines in $\mathbb{R}^2$. We started with lines in the plane and found that they correspond to planes through the origin $ax + by + cz = 0$. Hence, we may think of *defining* lines in projective space as being planes through the origin. Now, the plane $z = 0$ will never intersect our plane $z = 1$ (indeed, that would give the equation $1 = 0$, which is nonsensical). However, recall that the points that have z -coordinate 0 represent our points at infinity. Since $z = 0$ contains all of these points, we can consider the plane $z = 0$ to correspond to a line at infinity. You can imagine this as sort of being an infinitely large circle around our plane $\mathbb{R}^2$ (with the technicality that opposite points are identified as being the same). This follows a somewhat common theme in math of noticing something that matches an idea that we know but is more general, redefining our notions based on this more general idea, and seeing what conclusions we can come to.

We can also understand the intersection of lines in $\mathbb{R}^2$ by looking at the planes they represent in 3D. Two distinct lines in the plane (usually) intersect at a point, while two distinct planes through the origin always intersect at a line. This line of intersection corresponds to the point of intersection. To be more concrete, let's say we're considering the lines ℓ, m in $\mathbb{R}^2$. If they're corresponding planes are L and M , then the intersection of L, M would be some line P that goes through the origin in 3D. The line P then corresponds to the point p which is the intersection of ℓ and m .

One interesting consequence of this is that parallel lines actually do intersect! Two planes that correspond to parallel lines will intersect on a line in 3D, but this line will also lie in the plane $z = 0$ (try convincing yourself of this). This line then corresponds to a point at infinity in $\mathbb{R}P^2$. Hence, the two lines now intersect "infinitely far away." This is the sense in which points at infinity correspond to slopes. Every line with a certain slope goes through a point at infinity corresponding uniquely to that slope, and any two such parallel lines will intersect at this point at infinity. You can check that the slope is even what you want it to be. Check that the plane corresponding to any line of the form $y = \frac{1}{2}x + c$ for c any real number goes through the point at infinity $[1 : 1/2 : 0]$.

4 Duality of Points and Lines

In 3D, we have the property that every plane has "one perpendicular direction." It should be quite clear intuitively what is meant by this: the vertical direction is perpendicular to a table surface for example. Since directions are best represented as vectors, we can say that every plane has a perpendicular vector (up to scaling). The importance of this is that given a plane through the origin, we can find a unique line through the origin that is perpendicular to the plane. This essentially corresponds to a strong relation between the plane's corresponding line in 2D with the line's corresponding point which we will explore and exploit.

More formally, a vector is perpendicular to a plane if it is perpendicular to every vector lying on that plane. Now, we have represented points as lines which we have thought of as vectors that can be scaled, while we have thought of planes using their equations. So what is the relation between a plane and its perpendicular vector?

Theorem 4.1. *The vector (a, b, c) is normal to the plane $ax + by + cz = 0$.*

Since we really only care about planes that go through the origin, I set the right hand side of the plane to 0 but I encourage you to repeat the proof with the right side as being some constant, d . What does changing the constant mean geometrically?

Proof. We aim to prove that $v = (a, b, c)$ is normal to every vector w that lies in the plane $ax + by + cz = 0$. A vector in the plane must start at some point (x_0, y_0, z_0) and end at some other point (x_1, y_1, z_1) which are both on the plane so that the vector can be written as $w = (x_1 - x_0, y_1 - y_0, z_1 - z_0)$. The triples

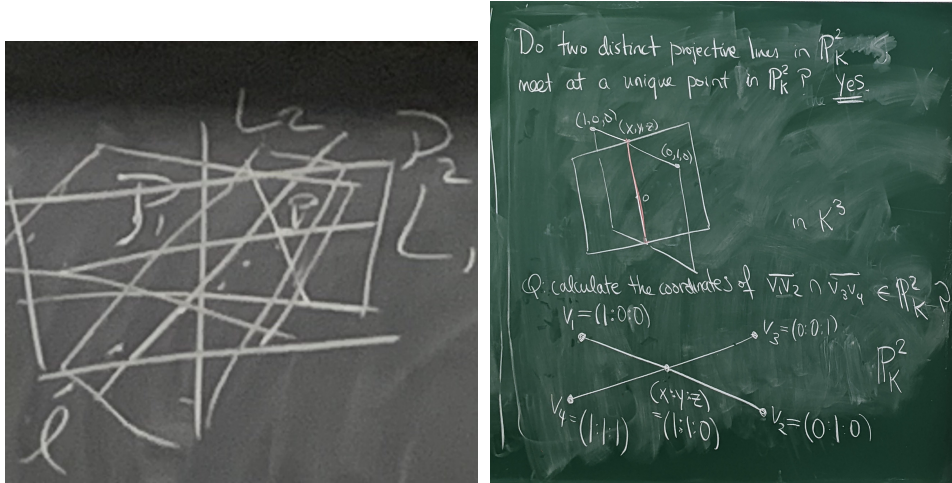


Figure 1: My attempt to draw the construction for Cramer's rule (left) and a well drawn diagram of the same construction by Melody Chan in lecture (right)

must both satisfy $ax_0 + by_0 + cz_0 = 0$ and $ax_1 + by_1 + cz_1 = 0$ (this is because this is the defining feature of a point lying on a plane). Now, we just need to show that $v \cdot w = 0$ since if the dot product of two vectors is 0, they are normal.

This is quite and direct.

$$v \cdot w = a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0) = (ax_1 + by_1 + cz_1) - (ax_0 + by_0 + cz_0) = 0$$

where in the last equality, we used the two equations that come from the points lying on the plane. $\square$

Okay, so now we know that the line normal to the plane $ax + by + cz = 0$ is $[a : b : c]$. In two dimensions, the plane corresponds to the line $ax + by + c = 0$ (substituting $z = 1$) while the line corresponds to the point $(a/c, b/c)$ (scaling our coordinates to $[a/c : b/c : 1]$ which lies on $z = 1$). Let's look at some consequences of these facts.

4.1 Cramer's Rule

The concept of duality can be used to prove Cramer's rule which is a statement of solving systems of equations in 2D using determinants. Here's the idea: say we have two lines ℓ, m in 2D and their corresponding planes are L, M . The point of intersection of ℓ, m can be denoted p with corresponding 3D line P . Now, P is a line that lies on L and M and is hence represented by a vector which lies on both planes. Now, we use the idea that a vector normal to a plane is normal to every vector on that plane. Let's denote the vectors normal to L, M as L', M' respectively. Hence, L', M' are two vectors which are both normal to P . Refer to figure 1 for a visual. L', M' are omitted from the drawing on the right, but you can imagine the two lines perpendicular to each plane.

Now, L', M' are distinct vectors (what would happen if they weren't?). Hence, all we need to do is find a vector normal to L', M' , and that vector will represent our line P . This can be done with the cross product!

As an example, consider the lines $7x + 3y + 31 = 0$ and $2x + 4y - 13 = 0$. The corresponding planes are $7x + 3y + 31z = 0$ and $2x + 4y - 13z = 0$ which have normal vectors $(7, 3, 31)$ and $(2, 4, -13)$ respectively. Now, the line common to both planes (which corresponds to the point of intersection) is

determined by the vector normal to both normal vectors i.e.

$$(7, 3, 31) \times (2, 4, -13) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 3 & 31 \\ 2 & 4 & -13 \end{vmatrix} = (-39 - 124)\hat{i} - (-91 - 62)\hat{j} + (28 - 6)\hat{k} = (-163, 153, 22).$$

The point of intersection hence corresponds to the homogenous coordinates $[-163 : 153 : 22] = [-163/22 : 153/22 : 1]$ (where the latter lies on the $z = 1$ plane). Hence, the point of intersection is $(163/22, 153/22)$. Even though the calculations may have been a bit hairy, they are still much cleaner than trying to solve using something like elimination or substitution.

Notice that it doesn't really matter which vector we take first in the cross product. Why? $v \times w = -w \times v$ but in projective geometry, we don't care about scaling by a factor (which is -1 in this case) so the order of the cross product will not change our final point.

Now, a couple of things to think about. Consider these as exercises. There are two ways that two lines do not intersect at a point. The first is that they are distinct parallel lines. We discussed earlier that these should intersect at a point at infinity i.e. the vector representing our line should have z -coordinate 0. Take two such lines and compute the cross product. Do you get what you expected? What is the relation of the resultant vector's coordinates to the equation/slope of your lines?

The second issue is if both lines are the same. Do out the cross product for such an example (you can either scale up one equation of a line or literally use the same equation twice). What goes wrong? In my opinion, this is quite a powerful example since the answer is that there are infinitely many points of intersection, but when math doesn't know how to handle it, it returns something weird, almost like an error statement in code.

4.2 Cramer's Rule for Points

Even if you have heard of Cramer's rule, I am willing to bet (not actually, I don't have money) that you have not seen this formula before. It is an example of the principle of duality between points and lines: we can use the same method of finding the intersection of two lines to find the line that goes through two points. The logic is actually technically easier this way, but I wanted to present the regular Cramer's rule first since it's more common.

What is the idea? Take two points p, q with corresponding 3D lines P, Q . The line ℓ which contains both points corresponds to the plane L which contains both lines P, Q . The line L' that's normal to L is also normal to P, Q since P, Q lie on L . Now, we just need to compute L' .

Now, say that p has coordinates (x_0, y_0) and q has coordinates (x_1, y_1) . The lines P, Q then have vectors $(x_0, y_0, 1), (x_1, y_1, 1)$ respectively (since p, q lie on the plane $z = 1$). The cross product of these two vectors now gives the vector that represents L' as some (a, b, c) which then corresponds to the line of ℓ being $ax + by + c = 0$.

As an example, consider the line that passes through the points $(8, 13)$ and $(17, -21)$. In projective coordinates, these points become $[8 : 13 : 1]$ and $[17 : -21 : 1]$. The vector normal to both of these is

$$(8, 13, 1) \times (17, -21, 1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & 13 & 1 \\ 17 & -21 & 1 \end{vmatrix} = 34\hat{i} - (-9)\hat{j} - 389\hat{k}$$

so the plane is simply $34x + 9y - 389 = 0$.

Some similar exercises to the previous section. Take some point p and choose some point at infinity (any vector with z -coordinate 0 e.g. $(1, 2, 0)$). Take their cross product and write the corresponding line in standard form. Does your line go through p as expected? What is the relation between the point at infinity and the slope of your new line ℓ ? What happens if you take two points at infinity? This should make the concept of points and the line at infinity a little more concrete and make it feel like that they do indeed exist in 2D in some strange sense.

Now, say you take the same point twice. What is the line that goes through them through this cross product method? Of course the actual answer is that there are infinitely many such lines since you can just choose any line that goes through the point. This sort of gives the same "error message" feel that we had in the previous section.

4.3 Projection of a Point onto a Line

This is a bit more exotic and pretty example. One relatively common question in 2D is what is the point q on a line ℓ which is closest to some other point p not lying on ℓ . Let's go through this step by step in 2D and then translate these steps in our projective geometry setting.

First, we find the line m that goes through p and is perpendicular to ℓ . The intersection of ℓ with m gives you the desired point q . If you have not seen this before, try proving it by taking another line m' through p and making a right triangle to show that the intersection of ℓ and m' is further from p than the intersection of ℓ and m .

Now, these steps seem fairly simple but the computation can sometimes be a pain to calculate. Luckily, projective geometry essentially translates these steps directly quite nicely. If the equation of ℓ is $ax + by + c = 0$, let's denote the vector (a, b, c) as L . Now, let p correspond to the line P in 3D. If you did the exercises in the previous section, you will remember that the point at infinity corresponding to $(a, b, 0)$ lies on every line that is perpendicular to ℓ . Let's denote $(a, b, 0)$ as L' . Now, the line m normal to ℓ going through p also goes through the point corresponding to L' . Hence, we can use our extended Cramer's rule for points to compute m since it just corresponds to $M = L' \times P$. Now, the intersection of ℓ and m is represented by $L \times M$ using the usual Cramer's rule. Hence, our point q in question is simply the point corresponding to $Q = L \times (L' \times P)$ where L' is the same as L with the third coordinate set to 0.

That feels really concise but you may be put off by having to compute a cross product twice. However, if you know a good amount about vectors and can see what's coming, you should be getting somewhat excited. We can actually get rid of all of the cross products by using the following formula:

Theorem 4.2 (Vector Triple Product). *For vectors A, B, C , $A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$.*

With this tool, we can rewrite our formula for our point q . We can now write

$$Q = L \times (L' \times P) = (L \cdot P)L' - (L \cdot L')P.$$

That's all! We reduced the process of finding a perpendicular line and a point of intersection to computing two dot products, scaling two vectors, and taking their difference. Try out some examples. A couple more things to think about: What if our line is the line at infinity? What if the point already lies on the line?

4.4 Invariant Lines

Here is another application of thinking of lines projectively: given an affine transformation (linear transformations along with translations, or equivalently anything that takes lines to lines), what are the invariant lines. For example, say we flip the x and y axes. You can convince yourself that the lines $y = x$ and $y = -x$ will be left invariant. For a more interesting example, say we scale the plane by a factor of 2 and then translate all points by the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. It's difficult to immediately think of what lines would be invariant by this transformation, or if there would even be any at all. To think of how to solve this problem, we can first think of finding things that are invariant under a linear transformation. For example, let's say a matrix A has an eigenvector v . Then, because A simply scales v , the line through the origin in the direction of v is left invariant by A . For example, $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ has an eigenvector $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Try checking for yourself that any point on the line $y = x$ is sent to another

point on $y = x$. So you may think that all invariant lines of a linear transformation are just the lines through the origin in the direction of an eigenvector. In contrast, consider the line $y = x + 4$. Check that taking points on this line and acting on it by the matrix A will produce another point on this line. This means that even without introducing the translations of affine transformations, we can still get invariant lines that don't go through the origin!

One way to try and solve this problem is to set a line as $y = mx + b$ with m, b as unknowns and solve for m, b given that the transformation leaves it invariant. To do this, take the vectors $r = \begin{pmatrix} x \\ mx + b \end{pmatrix}$ and $R = \begin{pmatrix} X \\ mX + b \end{pmatrix}$ and act on it with the affine transformation to try and calculate m, b . In the above example,

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ mx + b \end{pmatrix} = \begin{pmatrix} X \\ mX + b \end{pmatrix}.$$

In effect, this means you take any point on your line $(x, mx + b)$ and send it to a potentially different point $(X, mX + b)$ on the same line. Here, we have 4 variables and 2 equations, so we expect to have 2 "degrees of freedom." One of these is just that x can be any input value (since you can choose any point on your line). The other one corresponds to potentially multiple solutions for m, b . To solve the above, we could start with $X = (m + 1)x + b$ and substitute that into the other equation we get from multiplying out the second row of the matrix. I encourage you to work through the above example in detail and determine exactly what lines are preserved. The key is that the matrix above holds for all x , so what should the coefficient of x be? If you were to add translation by v , you can do the same procedure except consider $Ar + v = R$.

Admittedly, the method above is really nice and arguably easier to work out than the following method. That being said, the above method has some drawbacks. First, it doesn't account for lines of the form $x = a$, which is easy because you can check those separately. Second, it certainly does not explain why some slopes allow for lines not at the origin while others don't. Lastly, the calculations are somewhat messy and it doesn't feel like really there's anything interesting about the problem. In the following method, we are able to write our lines more abstractly and come with a meaning for an invariant line of an affine transformation.

Let's deal with the first problem first. We've already seen that if we want to account for all lines, it is useful to somehow incorporate standard form and write our line as $ax + by + c = 0$. All of these notes are based on the theme of projective geometry, so of course we're going to move to the third dimension for convenience. Our line now becomes the plane $ax + by + cz = 0$ and points on our original line correspond to points of the form $(x, y, 1)$. Now that we have moved our lines to 3D, how do we move our transformation to 3D? For this, consider the transformation

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} a_{11} & a_{12} \\ a_{13} & a_{14} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = AX + v$$

which is the most general possible affine transformation. From this, consider the matrix

$$M = \begin{pmatrix} a_{11} & a_{12} & v_1 \\ a_{13} & a_{14} & v_2 \\ 0 & 0 & 1 \end{pmatrix}.$$

It can be easily verified that M acting on the vector $(x, y, 1)^T$ gives $(x', y', 1)^T$ where $A(x, y)^T + v = (x', y')^T$. This means that not only does our transformation preserve the $z = 1$ plane, but we can write any affine transformation using some suitable M .

We are finally ready to tackle our problem of invariant lines under an affine transformation, which with the machinery we have, is actually quite nice. Let's rewrite our lines as

$$ax + by + cz = 0 \rightarrow \begin{pmatrix} a & b & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = AX = 0$$

where A is the row vector (a, b, c) and X is the column vector $(x, y, z)^T$. After the transformation M , X is sent to MX . So if the new point still lies on the line, $A(MX) = 0$. So the condition for an invariant line simplifies to the following statement: the invariant line to an affine transformation M corresponds to a row vector A such that whenever $AX = 0$, $AMX = 0$. Now, since $AX = AMX = 0$, we can combine them so that $AMX - \lambda AX = 0$ for any real number λ (its purpose will become clear later). Rewriting this, $(AM - \lambda A)X = 0$. This looks very similar to an eigenvector equation. We could hope that any invariant line satisfies $AM - \lambda A = A(M - I\lambda) = 0$ i.e. A is left eigenvector for some eigenvalue λ . In fact, this is true!

Now, the process of calculating left eigenvectors is essentially the same as solving for normal eigenvectors: solve for the eigenvalues through the characteristic polynomial and find the (left) null space of $M - \lambda I$. There is a much easier method however that I prefer: use Wolfram Alpha.

Let's go back to our previous example. The matrix M is written as

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The left eigenvectors can be found and turn out to be

$$(0, 1, 0), (-1, 1, 0), (0, 0, 1).$$

The first two correspond to the lines $y = 0$ and $y = x$ (which should be the same that you found earlier). What does the third correspond to? Well it's $z = 0$ which is the line at infinity. Of course any affine transformation keeps things that are infinitely far away beforehand infinitely far away afterwards. There's still the question of what gives with the solutions $y = x + b$ for any b . Well there's one piece of information I hid: the eigenvalues of the eigenvectors above. The first has eigenvalue 2 while the latter two have eigenvalue 1. In general, the sum (more generally, linear combination) of two eigenvectors is not an eigenvector *unless* the eigenvectors have the same eigenvalue. In this case, the last two vectors form an eigenspace of eigenvalue 1 so any vector of the form $a(-1, 1, 0) + b(0, 0, 1) = (-a, a, b)$ is an eigenvector of eigenvalue 1. These correspond to the lines $y = x + b$! This is interesting because I like to think of it as "the line at infinity is seeping into finite space" to produce more invariant lines. Even more exotic results come when two eigenvectors separate from $(0, 0, 1)$ (which is always an eigenvector) have the same eigenvalue (one example is given in the exercises below).

Some notes on this section. We've been saying that lines are dual to points. If you know some linear algebra, you know that the dual space to a vector space is the vector space of linear transformations to the field of scalars. By writing lines as $AX = 0$, we found a one-to-one (modulo scaling) correspondence between linear functionals on $\mathbb{R}^3$ and lines on $\mathbb{R}^2$ (or actually, $\mathbb{R}P^2$). So the space of lines really is like the dual space of points (if you consider points in $\mathbb{R}^3$, up to scaling). The way we solved this problem was by considering a transformation M that usually acts on points as $X \rightarrow MX$ and looking at its induced action on the dual space i.e. $A \rightarrow AM$ (the space of lines). Invariant lines were then just elements in the dual space invariant (up to scaling) under this transformation.

Here are some exercises. Show that if we have a transformation $X \rightarrow AX + v$ and we construct the matrix M from A, v as above, then the characteristic polynomial for M is that of A multiplied by an extra factor of $(\lambda - 1)$. In other words, show that the eigenvalues of M are the same as that of A with an extra eigenvalue of 1. Show that the left eigenvector corresponding to the extra 1 is $(0, 0, 1)$. This just means that any affine transformation fixes the line at infinity.

Consider one of the early examples of scaling by two then translating by $(1, 2)^T$ with transformation matrix given by

$$M = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find all eigenvectors and eigenvalues. Based on this, characterize all lines invariant by this affine transformation. Try taking some points on these lines and verify that they remain on the line after

the transformation. For kicks, try computing these lines through the previous method of solving via something of the form $Ar + v = R$.

5 Dual Curves

The idea of a dual curve generalizes the duality of lines and points and is quite remarkable that it even works. The idea is that for any given curve γ in the plane (that's nice enough, to be discussed later), there is another curve in the plane $\bar{\gamma}$ which I will denote $\text{dual}(\gamma)$. The duality function which takes in a curve and gives its dual curve is its own inverse i.e. $\text{dual}(\text{dual}(\gamma)) = \gamma$. I will assume this for now, but will come back to prove it very elegantly later on when we have an explicit way to represent the dual curve.

Now one issue we had to deal with earlier was how to represent geometric objects. For example, lines in standard form were much better suited for our needs. Now for curves, we will define them parametrically instead of through an equation. For example, if γ denotes the unit circle, we will represent it as $\gamma(t) = (\cos(t), \sin(t))$ rather than as $x^2 + y^2 = 1$. Some motivation for this: we know how to deal with points and lines, but not curvy lines. An equation gives some curve while using a parametrization allows us to focus on one point at a time, which we know how to handle.

Okay enough suspense, what is the dual curve? To make one, you fix some t_0 and consider the point $p = \gamma(t_0)$ on your curve γ . Now take the tangent line to γ at p and take the dual point to that line q_{t_0} . Now repeat the process for all values of t . The collection of points q_t will form your dual curve.

Let's walk through an example. Consider the curve $\gamma(t) : (t, t^2)$. This parametrically defines the parabola $y = x^2$. Now the tangent at the point (t, t^2) is $y - t^2 = 2t(x - t)$. Putting the line into standard form, $2tx - y - t^2 = 0$ giving the dual point $(-2t^{-1}, t^{-2})$. Notice that this depends on the parameter t so we already have a parametric description of the dual curve $\text{dual}(\gamma) : (-2t^{-1}, t^{-2})$. This is just the equation for $y = x^2/4$ if you prefer.

Let's try and look at how to do this process in general. The difficult part is calculating the tangent line. But since we have a parametric curve, we can use the rule

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}.$$

This rule follows from the chain rule applied to $\frac{dy}{dt}$ with x as an intermediate variable.

We will parametrize the curve as $\gamma(t) = (f(t), g(t))$ for some differentiable functions f, g . Now, for some t_0 , the point in consideration is $(f(t_0), g(t_0))$ and the slope is $\frac{dy}{dx}$ (remembering to evaluate it at t_0). The equation of the tangent line can be found by slope point form:

$$y - g(t_0) = \frac{dy}{dx}(x - f(t_0))$$

and if we use the rule from above, $\frac{dy}{dx} = \frac{g'(t_0)}{f'(t_0)}$, so with some manipulation, the line becomes

$$f'(t_0)(y - g(t_0)) = g'(t_0)(x - f(t_0))$$

but since we like standard form, we can convert this to

$$-g'(t_0)x + f'(t_0)y + (f(t_0)g'(t_0) - f'(t_0)g(t_0)) = 0.$$

Now, since the value of t_0 doesn't matter, we can consider this tangent line to vary with the parameter t . Finally, taking the dual point to each tangent line, we arrive at a parametrization for the dual curve:

$$\text{dual}(\gamma) = \left(-\frac{g'(t)}{f(t)g'(t) - f'(t)g(t)}, \frac{f'(t)}{f(t)g'(t) - f'(t)g(t)} \right)$$

And there you have it, the equation for a parametrization of the dual curve. Now, currently, it's in a very messy form and it is not at all obvious that the dual should be an involution (its own inverse), especially since $\text{dual}(\text{dual}(\gamma))$ would involve second derivatives of f, g in the calculation. As it has been with our earlier formulas in projective geometry, we will clean this up a bit.

The first step is to put γ in homogenous coordinates. For this, we just define $\gamma : [f(t) : g(t) : 1]$. Now, the problem of the dual curve becomes this: take the tangent line for some t , find the plane of that tangent line, and take the line normal to that plane in $\mathbb{R}^3$. As we have seen earlier, an efficient way to find the line normal to a plane is using the cross product. Now we need a second point on this tangent line. We can think of the 2D vector $\gamma' = \frac{d\gamma}{dt}$ as being the tangent vector to our curve, and so a second point is given by $\gamma + \gamma'$. We can extend this to taking a derivative in homogenous coordinates, since the third coordinate is just 1 so that $\gamma' = [f'(t), g'(t), 0]$. Now, we want to compute $\gamma \times (\gamma + \gamma')$, but because of linearity and anticommutativity, this just gives us $\gamma \times \gamma'$:

$$\text{dual}(\gamma) = [f(t) : g(t) : 1] \times [f'(t) : g'(t) : 0] = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ f(t) & g(t) & 1 \\ f'(t) & g'(t) & 0 \end{vmatrix}$$

and it can (should) be easily verified that this cross product gives the same description as the dual curve above, but in projective coordinates.

Now here is where things start to get nice. I will write the above relation as $\text{dual}(\gamma) = \gamma \times \gamma'$. Notice the significance of $[f'(t) : g'(t) : 0]$ as a point at infinity. We have used points at infinity as representing slopes, so what we're doing here is taking the cross product of γ , which encodes a point on the curve, and γ' , which encodes the slope of the curve at that point, everything we need to define a tangent line! One complaint that you may have is that when we move to projective coordinates, we don't want to restrict ourselves to having a third coordinate of 1. I leave it to you to verify that if we instead describe our curve with arbitrary homogenous coordinates, $\gamma = [f(t) : g(t) : h(t)]$, then $\gamma \times \gamma'$ still gives the tangent line when everything is projected to $z = 1$.

Now, let's finally get to proving that duality is an involution (self-inverse). The proof will rely on the following theorem:

Theorem 5.1 (Product Rule for Cross Products). $(v \times w)' := \frac{d}{dt}(v \times w) = v' \times w + v \times w'$.

It can be proved by expanding the cross product in terms of products of the components of each vector and using the product rule on each component. The proof is really quite clean from here and only uses the product rule and the triple product formula we came across earlier. Let $\bar{\gamma} = \gamma \times \gamma'$. Our goal is to prove $\bar{\gamma} \times \bar{\gamma}' \propto \gamma$ where we only care about it being proportional for each time t since as we are working projectively, we don't care about scaling.

$$\begin{aligned} \bar{\gamma} \times \bar{\gamma}' &= \bar{\gamma} \times (\gamma \times \gamma')' \\ &= \bar{\gamma} \times (\gamma' \times \gamma' + \gamma \times \gamma'') \\ &= \bar{\gamma} \times (0 + \gamma \times \gamma'') \\ &= (\bar{\gamma} \cdot \gamma'')\gamma + (\bar{\gamma} \cdot \gamma)\gamma'' \\ &= (\bar{\gamma} \cdot \gamma'')\gamma \propto \gamma. \end{aligned}$$

Let's describe the steps. First, we wrote $\bar{\gamma}$ as $\gamma \times \gamma'$ and applied the derivative product rule. The first cross product is 0 since it's γ' crossed with itself. Next, we applied the vector triple product and expanded in terms of γ and γ'' . The coefficient of γ'' is 0 because $\bar{\gamma} = \gamma \times \gamma'$ is perpendicular to γ , hence giving us some parametric curve proportional to γ . Now we have completed a relatively simple proof that duality is really an involution and the dual of the dual gives you back your original curve!

5.1 Envelopes

Now, let's discuss a cute little application of duality: finding envelopes. Firstly, what is an envelope? You have likely seen "optical illusions" of a bunch of straight lines that together, seem to trace out a

curve. That is essentially the concept of an envelope. If you have a collection of lines defined by some parameter t , the envelope of those lines is a curve that has all those lines as tangent lines. You can also think of this as the opposite of finding lines of tangency of a curve. That is, if you are given the lines of tangency, can you reverse engineer the curve?

Let's give an example of an envelope to make the idea more concrete. We return to our friend the unit circle $\gamma : (\cos(t), \sin(t))$. Using the formula from the previous section, it can be easily verified that the tangent lines are $\cos(t)x + \sin(t)y - 1 = 0$. So if our collection of lines is $\cos(t)x + \sin(t)y - 1 = 0$, the envelope is the unit circle. Check this on a graphing calculator like Desmos: graph the lines of the form above for various values of t and you will see the circle come into view.

If our lines are defined by $f(t)x + g(t)y + h(t) = 0$, we can do the following procedure to retrieve the original curve, exploiting the fact that the dual of a dual curve is the original:

- We have the tangent lines to our envelope γ
- We can take the dual points to each tangent line to obtain the dual curve $\text{dual}(\gamma) = \bar{\gamma}$
- Now, taking the dual of the dual, we get $\text{dual}(\bar{\gamma}) = \gamma$, giving us our envelope

As an example, consider the lines $2tx + 3y + t^2 = 0$. Now, we could take the dual points by dividing through by t^2 , but remember that duality doesn't care about scaling so we can use homogenous coordinates, which are the same as the coefficients of our lines. This means that we can consider our dual point as $\bar{\gamma}(t) = [2t : 3 : t^2]$. Now, we can take the dual to this:

$$\bar{\gamma} \times \bar{\gamma}' = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2t & 3 & t^2 \\ 2 & 0 & 2t \end{vmatrix} = 6t\hat{i} - 2t^2\hat{j} - 6\hat{k}$$

which means that our envelope is parametrized by $(-t, t^2/3)$ i.e. the curve $y = x^2/3$. Again, check it on Desmos. I recommend you to check that the tangent lines of $y = x^2/3$ are of the form above.

Now, let's get to the computational method of finding an envelope. We can use a parametrization to describe our lines that of tangency. That is, we use $\bar{\gamma}(t) = [f(t) : g(t) : h(t)]$ to denote the set of lines $f(t)x + g(t)y + h(t) = 0$. Now, if you noticed, we technically don't even need the dual points since our lines already gave us our functions f, g, h . Hence, if we just take $\gamma = \text{dual}(\bar{\gamma})$, we get our envelope.

Now here, we have something quite powerful. The points of γ of course parametrizes our envelope, but the points of $\bar{\gamma}$ don't denote some other curve, but instead describe the lines tangent to our curve. This suggests the following: we're used to parametrizing a curve by describing its points at each value of t , but equivalently, we can parametrize a curve by describing its tangent lines at each t . Note that I said tangent *lines*. If one of our tangent lines is ℓ , we don't know what point on ℓ lies on the curve i.e. we don't know the *point* of tangency. It may make you a bit uncomfortable to parametrize a curve by looking at its tangent lines since you don't know where it's tangent when looking at each line individually. But I argue this: tangent lines give the same amount of "local information" as a point does, just a different type. When you look at a point in isolation, you don't know the tangent line at that point unless you look at nearby points.

This gives a deep new interpretation of duality. We can parametrize a curve by its points and denote that parametrization as γ or by its tangent lines and denote that as $\bar{\gamma}$. The relation between these two parametrizations is duality: $\bar{\gamma} = \text{dual}(\gamma)$ and $\gamma = \text{dual}(\bar{\gamma})$. Now, duality doesn't represent some other curve in the plane but instead represents how to convert between two equivalent descriptions of the same curve.